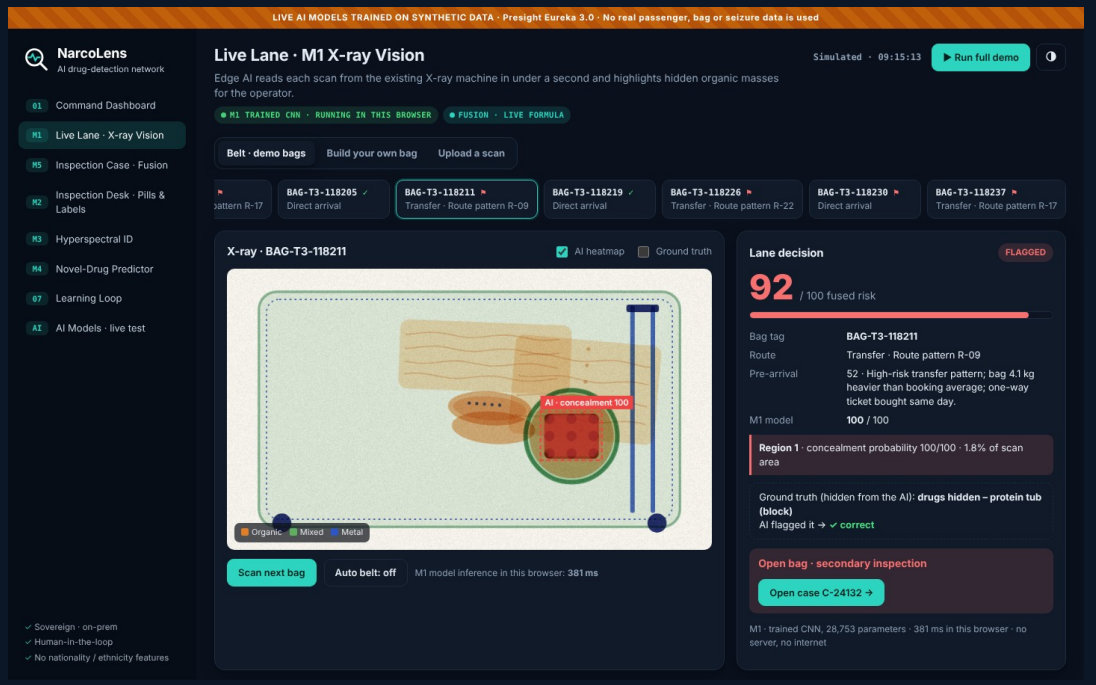
PRESIGHT EUREKA 3.0 · DEFENSE & POLICING

# NarcoLens

## Turning every UAE customs scanner into an AI drug detector

**Software-only** **Sovereign · on-prem** **Live AI models**

TRL 4 · Live AI models trained and tested on synthetic data, running in a clickable web prototype




*Prototype: M1 X-ray Vision flags a dense block hidden inside a protein tub*

# Smugglers are out-innovating manual screening

**1,273**

drug seizures at Dubai's land, sea and air ports in one year

**540**

drug cases at Dubai airports in nine months (Jan–Sep 2023)

**14M**

Captagon pills hidden in a single commercial shipment (Feb 2026)

*Figures from Dubai Customs and press reports, 2023–2026*

## Hidden in plain sight

- Pills pressed into clothing buttons
- Drugs packed inside furniture and door panels
- Opium mixed into protein powder and shoe linings

## Still found mostly by hand

- Operators judge X-ray density by eye, bag after bag
- Sniffer-dog teams cover limited hours and lanes
- Naming the substance waits days for the lab

# The gap: detection hasn't kept pace

## No AI reads drug X-rays

Operators judge density by eye. We found no published system that auto-flags narcotics at UAE checkpoints.

## Chemistry needs a lab

Confirming a substance takes days, so officers act on suspicion alone at the desk.

## Synthetic drugs mutate

New variants are designed to dodge detection libraries; detection lags behind the first seizure.

## Parcels are booming

E-commerce and "research peptide" shipments grow faster than inspection capacity.

**Why now:** edge GPUs and hyperspectral cameras are affordable, the UAE is investing in sovereign AI, and customs data is already digital. The missing piece is software that connects it all.

# NarcoLens: one AI layer, one case per bag

**M5 · Pre-arrival risk**

Route and booking patterns score each bag early

**M2 · Pills, vials, labels**

Imprint match, Arabic/English OCR, registry check

**M1 · X-ray Vision**

Edge AI reads the existing scan in under 1 s

**INSPECTION CASE**

one per bag tag
M5 fuses all evidence

**M3 · Hyperspectral ID**

Names the substance through the packaging

**Loop · Federated learning**

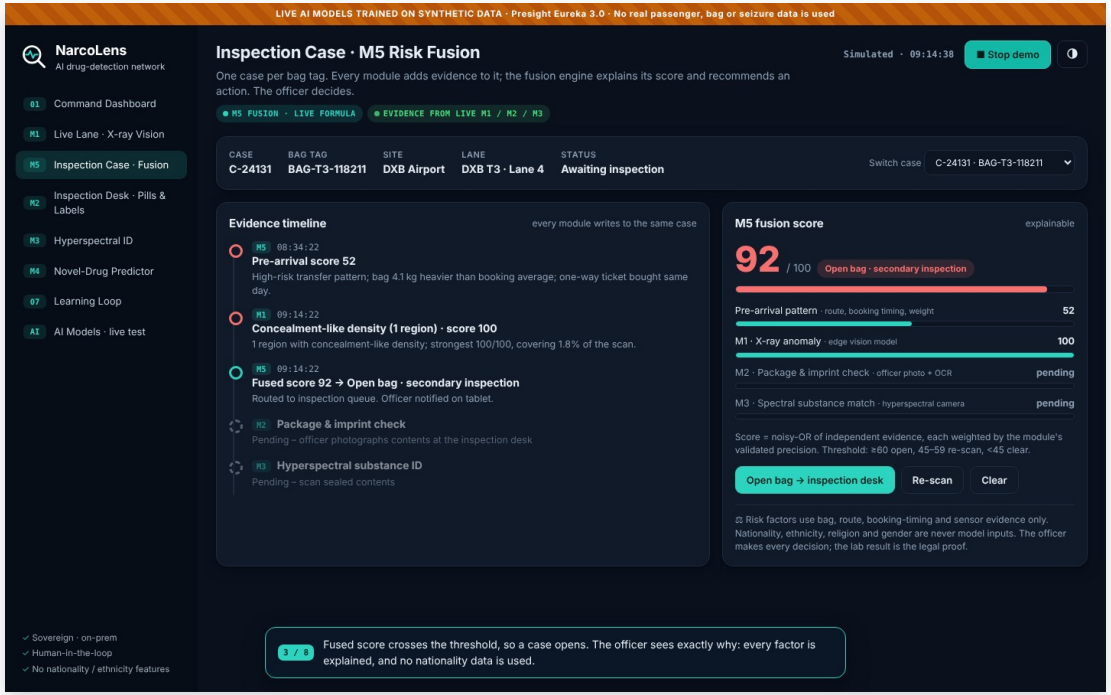
Each confirmed seizure retrains every site

**M4 · Novel-drug predictor**

Predicts new variants and feeds the M3 library

# A bag's journey through NarcoLens

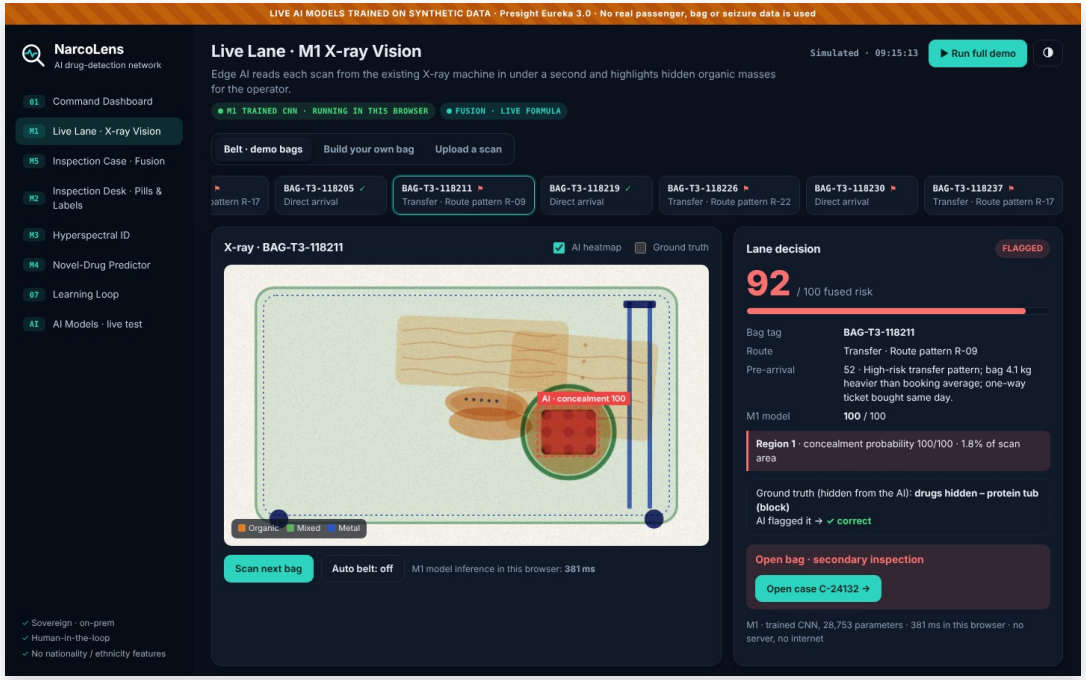
1. **Before landing · M5**
   Route and booking pattern score the bag 52/100
2. **On the belt · M1**
   Dense block in a protein tub flagged in 0.4 s (87)
3. **Case opened · M5**
   Fused score 83: open the bag, reasons shown
4. **Inspection desk · M2**
   Label not in registry; Captagon-style imprint
5. **Hyperspectral · M3**
   Amphetamine-type stimulant, 98% match, still sealed
6. **Lab confirms · Loop**
   Seizure confirmed; all 7 sites retrain, no images shared

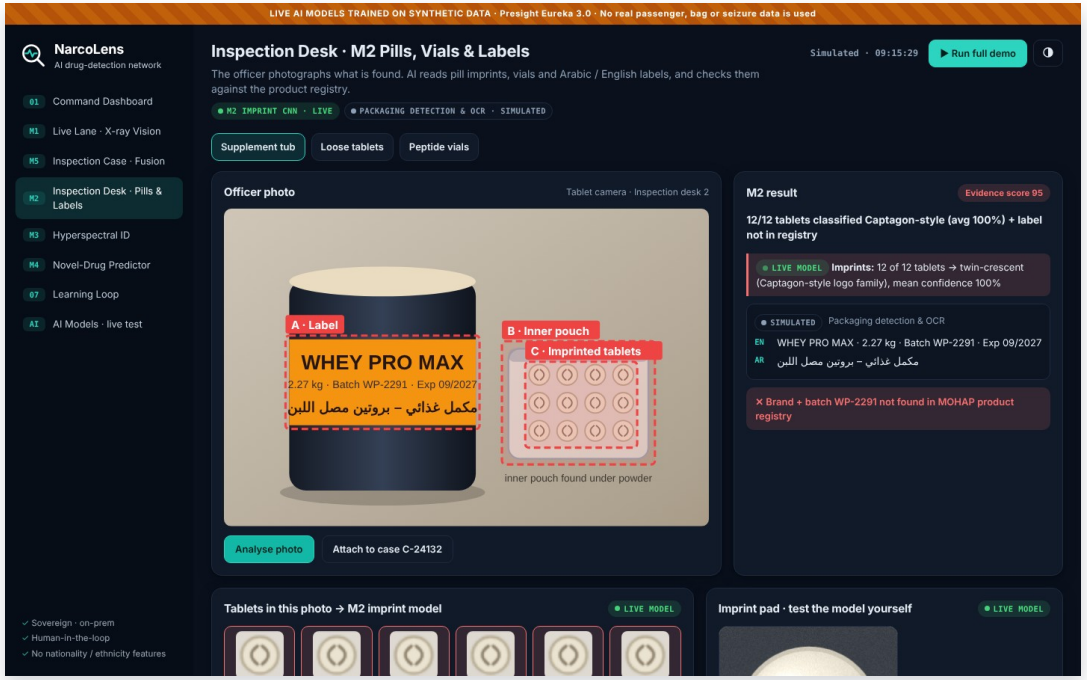



*Prototype: every module writes to the same case, and the score explains itself*

Under a minute from belt to substance ID in the demo, and no bag is opened until the evidence justifies it

# Seeing what's hidden: the vision modules



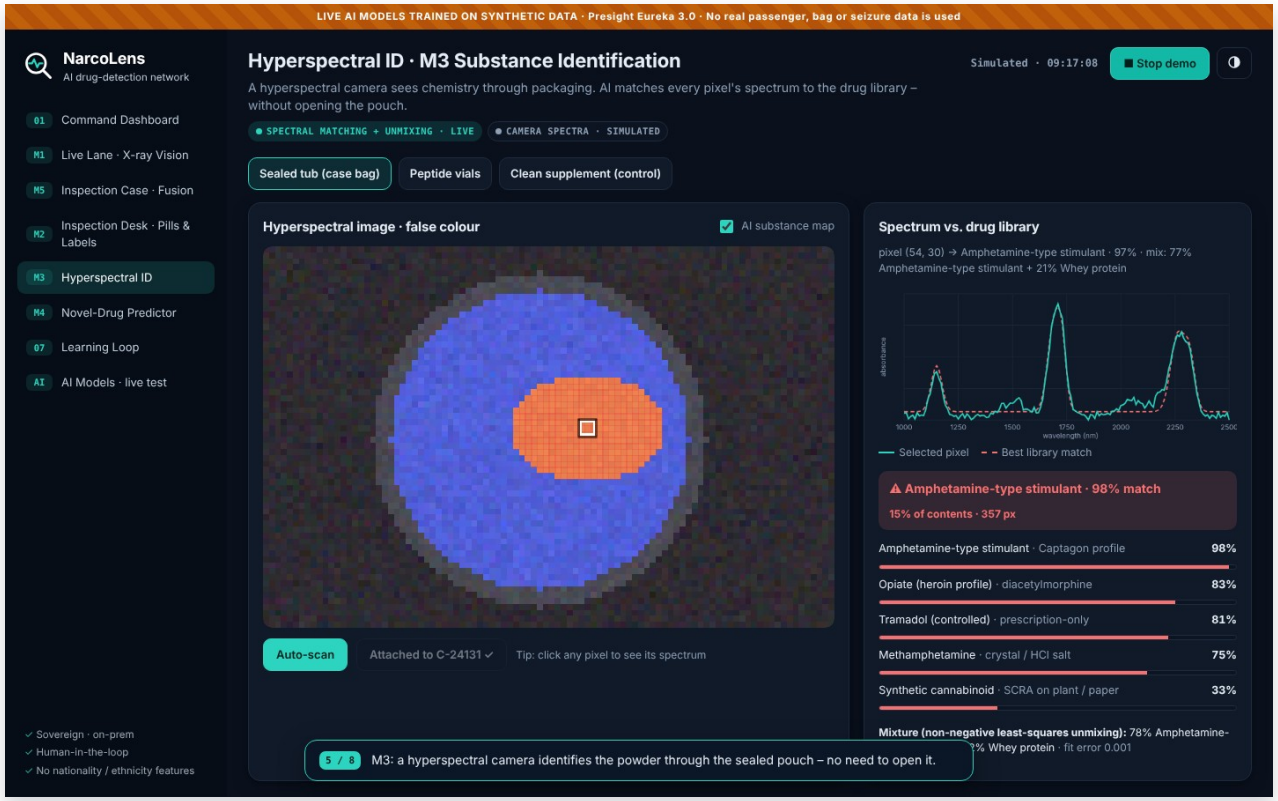

## M1 · X-ray Vision

- Edge GPU beside the existing scanner, under 1 s per bag
- Finds organic masses, hidden cavities, abnormal density
- Prototype: trained CNN, 100% recall on unseen bags

## M2 · Pills, vials and labels

- Officer photographs what is found on a desk tablet
- Matches pill imprints and vials; reads Arabic/English labels
- Prototype: imprint CNN, 99.8% accuracy; OCR next

# Seeing chemistry: identify, then predict




*Prototype: M3 finds amphetamine-type powder inside a sealed pouch (98% spectral match)*

## M3 · Hyperspectral ID

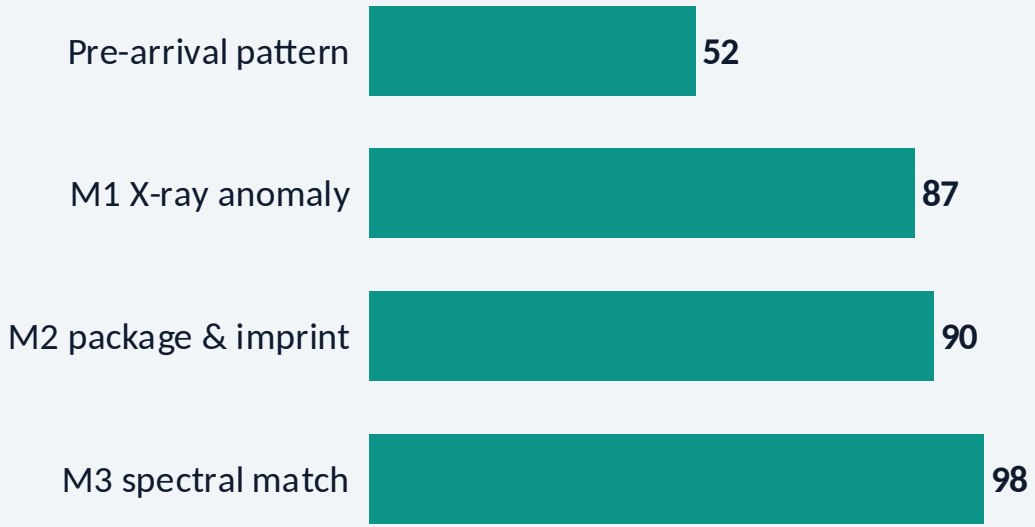
- Camera reads ~150 bands from 1,000 to 2,500 nm
- Every pixel matched to the drug library
- Names the drug without opening the package

## M4 · Novel-drug predictor

- Graph neural network predicts new variants' spectra
- Chemists validate, then it reaches every camera
- Recognises a variant before its first seizure

# Explainable by design, accountable by default

## M5 fusion: why this bag scored 99




Each bar is independent evidence, weighted by the module's validated precision. Thresholds: 60+ open · 45–59 re-scan · under 45 clear.

### No nationality or ethnicity

Only bag, route, timing and sensor evidence. Bias audits every quarter.

### The officer decides

AI recommends; every search and seizure is a logged human decision.

### The lab is the legal proof

AI evidence pack (images, spectra, scores) kept under chain of custody.

### On-prem and federated

Models train at each site; zero raw images leave a site.

# Software-only, vendor-agnostic architecture

## 1 · Edge

- GPU box beside each X-ray/CT scanner and hyperspectral camera
- Runs M1 and M3 locally in under 1 s
- Reads images via the DICOS standard

## 2 · Core (on-prem)

- Inspection cases and M5 fusion
- M4 predictor and drug library
- Federated aggregation, model registry, audit log

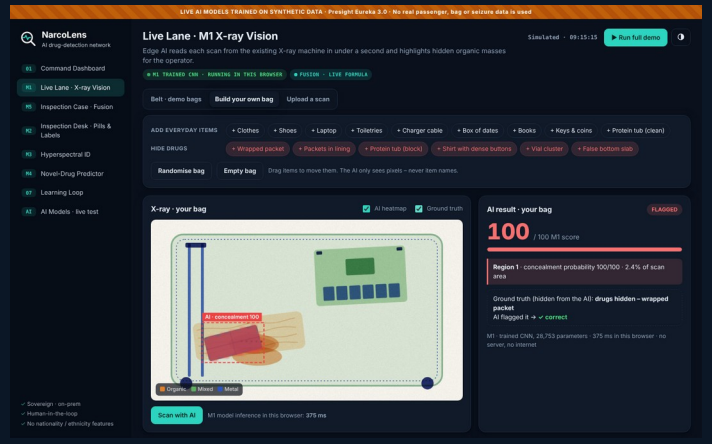
## 3 · Apps

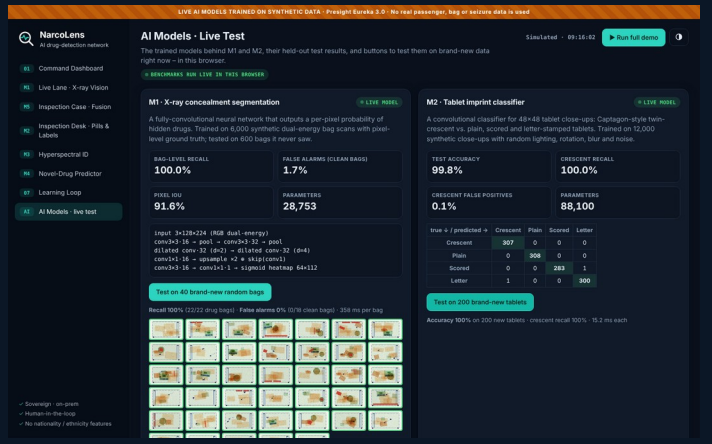
- Overlay on the scanner operator's screen
- Officer tablet app at the desk
- Command dashboard for supervisors

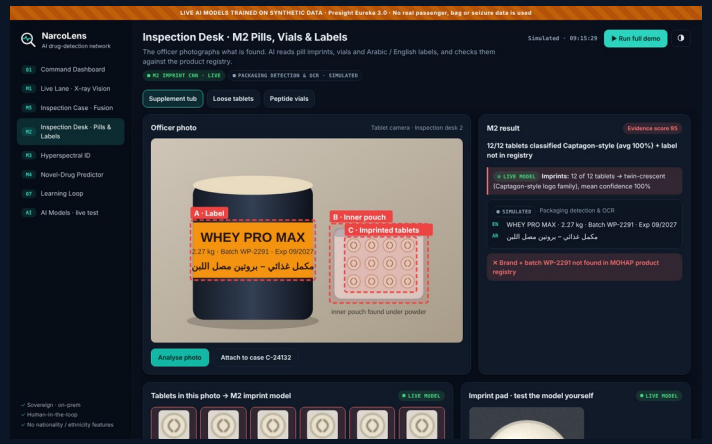
**Integrates with**
Any scanner vendor (DICOS) · customs manifest and booking systems · forensic lab systems

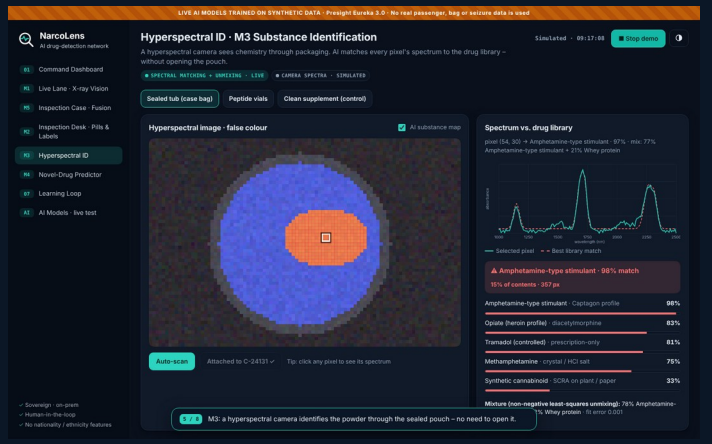
**Built with**
PyTorch · RT-DETR · PatchCore · ViT/CLIP · Arabic OCR · PyTorch Geometric · XGBoost · Flower · Kubernetes · React

# Working prototype with live AI models




*Bag Builder · AI model cards · Imprint model + pad · Spectral ID*

**100%** M1 recall on unseen bags

**1.7%** M1 false alarms, clean bags

**99.8%** M2 imprint accuracy

1. Build a bag, hide drugs anywhere, scan
2. Draw a pill imprint on the pad
3. Benchmark on brand-new random bags

**Trained on synthetic scans · runs offline in any browser**

# Expected impact: pilot targets

**+30%**

seizures per 1,000 bags inspected

**Minutes**

to identify a substance, vs ~3 days in the lab

**−40%**

operator review time per flagged bag

**≤5%**

extra false alarms per lane

*Targets to be validated in an 8-week shadow-mode pilot before any live use*

## Who benefits

- Dubai and Abu Dhabi Customs, ICP
- Police anti-narcotics units
- Honest travellers: fewer manual searches

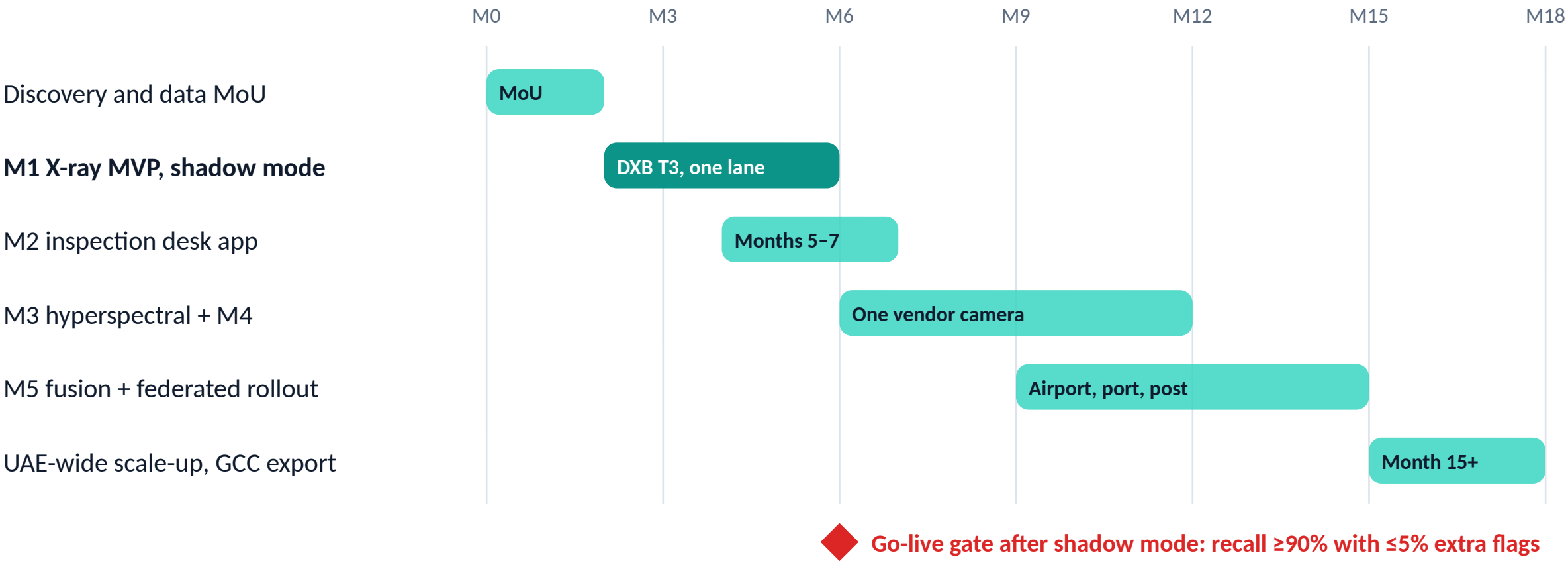
## Value for Presight

- Sovereign product for UAE border security
- Export to GCC customs agencies
- Recurring licence and support per site

## Beyond drugs

- Weapons and explosives
- Counterfeit goods and wildlife trafficking
- Undeclared currency and gold

# Roadmap: shadow-mode pilot to UAE-wide rollout




**Team at peak ≈ 12:** 3 computer-vision engineers · 1 spectroscopy ML scientist · 2 data engineers · 2 MLOps · 2 full-stack · product lead with customs expert · responsible-AI and legal lead

# Risks we've designed for

| Risk | How NarcoLens handles it |
| --- | --- |
| **Little labelled drug X-ray data** | Threat-image projection, public X-ray pretraining, active learning from officer feedback |
| **False alarms slow the queues** | Shadow mode first; per-lane thresholds; officers see only top-risk bags |
| **Closed scanner formats** | DICOS standard; early partnership with one or two scanner vendors |
| **Profiling and bias** | No nationality or ethnicity features; quarterly bias audits; full audit log |
| **Evidence in court** | AI only flags; the lab test stays the legal proof; explainable heatmaps and chain of custody |

## Why Presight

- Sovereign, on-prem AI for UAE government is our core business
- Big-data analytics for public safety and investigations
- In-house talent across computer vision, Arabic NLP and MLOps

# The team and the ask

## Team

[Name · Role · Department]

[Name · Role · Department]

[Name · Role · Department]

[Name · Role · Department]

[Name · Role · Department]

*Up to 5 Presight employees*

## Our ask

1. **An ELT sponsor**
   to champion NarcoLens with UAE border agencies
2. **An introduction to Dubai Customs and ICP**
   to agree a data-sharing MoU for historical scans
3. **A 3-month shadow-mode pilot**
   on one DXB Terminal 3 lane, with no impact on passengers

***One seizure at DXB should make every UAE border smarter.***